

Question 1

Which of the following is an advantage of a good corporate governance structure?

- A. Decline in the operational freedom and efficiency.
- B. Increase in business and investment risk.
- C. Decrease in the cost of debt.**

C is correct. An effective corporate governance structure makes it more likely that a company will meet its debt obligations and therefore reduces the cost the debt. Introduction to Corporate Governance and Other ESG Considerations; Section 7.2.

Question 2

Which of the following is most likely a part of the execution step of the portfolio management process?

- A. Portfolio construction.**
- B. Portfolio monitoring.
- C. Performance measurement.

A is correct. Portfolio construction is a part of the execution step of the portfolio management process. Portfolio Management Overview; Section 4.

Question 3

Which theory suggests that managers choose methods of financing that send the least signal to outsiders?

- A. Static Trade-off theory
- B. Stakeholder theory
- C. Pecking order theory**

C is correct. *Pecking order theory* suggests that managers choose methods of financing that send the least signal to outsiders. The preferred hierarchy for financing is:

- First preference: internal financing (retained earnings)
- Second preference: debt financing
- Third preference: equity financing

The *static trade-off theory* is based on balancing the expected costs from financial distress against the tax benefits of debt service payments.

Stakeholder theory is based on the premise that a company's focus is not restricted to shareholders, but extends to other stakeholders as well such as its customers, employees, suppliers, etc. It gives more importance to ESG considerations by making them a clear objective for the board and management. *Capital Structure*.

Question 4

ILC Co. raises 70 percent of its finances through debt whereas MUN Co. raises 30 percent of its finances through debt. ILC Co. most likely has:

- A. higher operating risk.
- B. higher financial risk.**
- C. higher sales risk.

B is correct. Financial risks increase with greater use of fixed cost financing. Measures of Leverage; Section 3.

Question 5

The NexCare Corp. has accounts payable of €1.5 million with terms of 1/10, net 35. Accounts receivable stands at €2 million. The company also has €1.5 million in marketable securities. NexCare has a short-term need of €300,000 to meet working capital requirements.

Which of the following options appears to be most suitable for raising the €300,000?

- A. The company should delay paying accounts payable and forgo the 1% discount.
- B. The company should sell some of its marketable securities at a 0.5% brokerage cost.**
- C. The company should sell some of its accounts receivable to a factor at a 5% discount.

B is correct. The options available for raising €300,000 are summarised below:

Liquidation Cost		Action	Source
€	%		
3,000	1	Delay payment of €300,000 and forgo 1% discount	Accounts payable (1/10, net 45)
15,789.5	5	Sell €315,789.5 in value at 5% discount to raise €300,000	Accounts Receivable
1,500	0.5	Sell 300,000 in value	Marketable securities

Sale of marketable securities is the least costly option for meeting working capital requirements of €300,000. Marketable securities are liquid instruments and can be easily sold for market value less the brokerage cost. *Sources of Capital*

Question 6

At the start of the year 2014, Daisy Merlin buys one share of a stock for USD14. At the start of 2015, Daisy buys another share of the same stock for USD13. The stock paid a dividend of USD0.50 per share each year. On January 1, 2016, Daisy receives a total of USD31.60 for selling the two shares. The time-weighted and the money weighted returns for Daisy are closest to:

Time weighted return & Money weighted return are

- A. 9.96% & 14.35%**
- B. 5.80% & 11.82%
- C. 9.88% & 17.28%

A is correct. To determine the time-weighted return, calculate the holding period return.

HPR:

$$\text{HPR}_1 = (13 + 0.5) / 14 - 1 = -3.57\%$$

$$\text{HPR}_2 = (31.6 + 0.5 \times 2) / (13 \times 2) - 1 = 25.38\%$$

$$\text{Time weighted return} = [(0.9643)(1.2538)]^{0.5} - 1 = 9.96\%$$

Using a financial calculator, compute IRR.

$$\text{CF}_0 = -14, \text{CF}_1 = -13 + 0.5 = -12.5, \text{CF}_2 = 31.6 + 0.5 \times 2 = 32.6, \text{CPT IRR} = 14.35\%.$$

Money weighted return = 14.35%. Portfolio Risk and Return Part I; Section 2.1.

Question 7

A company decides to charge high price for its products or removes the product safety features to reduce costs. Such a scenario can create conflict of interest between:

- A. customers and suppliers.
- B. customers and shareholders.**
- C. shareholders and creditors.

B is correct. When a company charges higher price for its products or reduces product safety features to reduce costs, it can create a conflict of interest between customers and shareholders. Introduction to Corporate Governance and Other ESG Considerations; Section 3.

Question 8

A “tag cloud” is a Big Data visualization technique where:

- A. words that appear less frequently are shown with a larger font and words that appear more often are displayed with a smaller font.
- B. the relationship between different concepts is shown.
- C. words are sized and displayed based on the frequency of the word in the data file.**

C is correct. A Big Data visualization technique applicable to textual data is a “tag cloud,” where words are sized and displayed on the basis of the frequency of the word in the data file. A is incorrect because words that appear more often are shown with a larger font, and words that appear less often are shown with a smaller font. B is incorrect because “mind map” another data visualization technique shows how different concepts are related to each other. Fintech; Section 5.2. LO.b.

Question 9

Consider the following two mutually exclusive projects:

Project Year 0 Year 1 Year 2 Year 3

Project	Year 0	Year 1	Year 2	Year 3
A	(7,653)	5,304	6,666	7,002
B	(5,020)	4,800	3,289	2,465

If the annual discount rate is 7 percent for both the projects, the firm should most likely accept:

- A. project A.**
- B. project B.
- C. both projects.

A is correct.

Project A: $CF_0 = -7,653$, $CF_1 = 5,304$, $CF_2 = 6,666$, $CF_3 = 7,002$, $I = 7\%$, CPT NPV, $NPV = 8,842$.

Project B: $CF_0 = -5,020$, $CF_1 = 4,800$, $CF_2 = 3,289$, $CF_3 = 2,465$, $I = 7\%$, CPT NPV, $NPV = 4,350$.

Since the two projects are mutually exclusive, the project with the higher NPV will be chosen, i.e. Project A. Uses of Capital

Question 10

Elephant Inc. has preferred stock that pays a dividend of USD15 per share. It currently sells for USD130 per share. Its cost of preferred stock is closest to:

- A. 19.5 percent.
- B. 8.7 percent.
- C. 11.5 percent.**

C is correct. $r_p = 15 / 130 = 0.115 * 100 = 11.5\%$. Cost of Capital; Section 3.2.

Question 11

An asset management firm generated the following annual returns in their Asian large cap equity portfolio:

Year	Net Return (%)
2010	12.4
2011	-22.2
2012	8.7
2013	-1.4

The 2014 return needed to achieve a trailing five-year geometric mean annualized return of 5.0% when calculated at the end of 2014 is closest to:

- A. 27.5%.
- B. 33.4%.
- C. 36.2%.**

C is correct. Holding period total return (cumulative) factor calculation through 2011:

$(1+0.124) \times (1-0.222) \times (1+0.087) \times (1-0.014) = 1.124 \times 0.778 \times 1.087 \times 0.986 = 0.9372\%$

Compound total return (cumulative) factor at 5% per year of five percent for five years:

$1.055 = 1.2763\%$

Return needed in 2014 to achieve a compound annualized return of 5%

$1.2763/0.9372 = 1.3618 = 36.2\%$

Check: $0.9372 \times 1.362 = 1.2763(1/5) = 1.050 = 5\%$ annualized. Portfolio Risk and Return Part I; Section 2.1.

Question 12

Which of the following statements about breakeven quantity is least likely correct ?

- A. The breakeven quantity is the level of sales at which net income is zero.
- B. The operating breakeven quantity is the level of sales at which EBIT is zero.
- C. The breakeven quantity is the level of sales at which net income and EBIT are equal.**

C is correct. The breakeven quantity is the level of sales at which net income is zero. The operating breakeven quantity is the level of sales at which EBIT is zero. Measures of Leverage; Section 3.6.

Question 13

Analyst 1: Straight voting and related party transactions create conflict of interest between controlling shareholders and minority shareholders.

Analyst 2: Straight voting and related party transactions create conflict of interest between managers and board of directors.

Which Analyst's statement is least likely correct?

- A. Analyst 1.
- B. Analyst 2.
- C. Both.**

B is correct. Straight voting and related party transactions create conflict of interest between controlling shareholders and minority shareholders. Introduction to Corporate Governance and Other ESG Considerations; Section 3.2.

Question 14

When considering capital projects, at times ranking conflict exists between NPV and IRR. In this context, which of the following is least accurate?

- A. Differing cash flow patterns can cause projects to rank differently.
- B. Project scale can lead to ranking conflicts between NPV and IRR.
- C. Non-conventional cash flow patterns can cause NPV values to rank projects incorrectly.**

C is correct. In case of non-conventional cash flows (more than one sign change in the cash flows), NPV value gives the correct decision compared to IRR which might have the multiple IRR problem. Uses of Capital

Question 15

Sara Smith, a technical analyst, observes the formation of a head shoulders pattern in a stock she has been following. Specifically, she notes the following information:

Head Price	\$37.50
Shoulder Price	\$31.20
Neckline Price	\$28.25

Her target price should be:

- A. USD19.00.**
- B. USD21.95.
- C. USD24.90.

A is correct. Price target = Neckline – (Head – Neckline)

Price target = $28.25 - (37.5 - 28.25) = 19.00$. Technical Analysis; Section 3.3.

Question 16

Enercon Enterprises is evaluating an investment in a £100 million project that will be financed with 50% debt and 50% equity. Management has already determined that the NPV of this project is £13 million if it uses internally generated equity. However, if the company uses external equity, it will incur flotation costs of 7.5%. Assuming flotation costs are not tax deductible, the NPV using external equity would be:

- A. less than £13 million because higher weighted average cost of capital would be used to incorporate the flotation costs.
- B. £13 million because flotation costs have no impact on NPV.
- C. £9.25 million after deduction of flotation costs.**

C is correct. Since the project will be financed with 50% equity, the company will issue £50 million of new stock. The flotation cost of external equity is $(0.075 \times 50,000,000) = 3,750,000$. The NPV of the project using external equity is 9,250,000 after deducting flotation cost from the NPV using internal equity. Adjusting the cost of capital to reflect the flotation cost is not a preferred way to account for flotation costs. *Cost of Capital – Foundational Topics*

Question 17

Which of the following is most likely considered a drag on liquidity?

- A. Bad debts.**
- B. Reduced credit limits.
- C. Making early payments.

A is correct. Drags on liquidity delay cash inflows, hence bad debts are drags on liquidity. *Sources of Capital*

Question 18

Given that the expected return on the market portfolio is 9 percent and the risk-free rate is 6 percent, the expected return of a security with a beta of 0.78 is closest to:

- A. 8.3 percent.**
- B. 17.7 percent.
- C. 11.3 percent.

A is correct.
 $E(R_i) = R_f + \beta(E(R_m) - R_f)$
 $E(R_i) = 6\% + 0.78(9\% - 6\%)$
 $E(R_i) = 8.34\%$. *Portfolio Risk and Return Part II; Section 4.*

Question 19

For capital budgeting purposes, flotation costs should be most likely:

- A. ignored.
- B. added to the cost of capital.
- C. subtracted as one of the project's initial cash flows.**

C is correct. For capital budgeting purposes, the correct treatment of flotation costs is that they should be considered as part of the project's initial cash flows. Ignoring flotation costs or adding them to the cost of capital are incorrect methods. Hence, A and B are incorrect. *Cost of Capital; Section 4.4.*

Question 20

The management of Prime Corp. has developed a new process that it believes produces a superior product. The company must make an initial investment of USD211 million to commence production. If demand is high, cash flows are expected to be USD40 million per year. If demand is low, cash flows will be only USD25 million per year. Management believes there is an equal chance that demand will be high or low. The investment also gives the company a

production-flexibility option allowing the company to add shifts at the end of the first year if demand turns out to be high. If the company exercises this option, net cash flows would increase by an incremental USD9 million in Years 2–10. Assume opportunity cost of funds is 9%.

NPV of the optimal set of investment decisions for Prime Corp. including the production-flexibility option will be *closest* to:

- A. \$22.32 million**
- B. \$24.55 million
- C. \$51.53 million

A is correct.

First, NPV without the production flexibility option will be calculated as follows:

If demand is “high,” the NPV is

$$NPV = -211 + \sum_{t=1}^{10} 140 / 1.09^t = 45.71$$

If demand is “low,” the NPV is

$$NPV = -211 + \sum_{t=1}^{10} 125 / 1.09^t = 50.56$$

The expected NPV without production flexibility option is $0.50(45.71) + 0.50(-50.56) = -\2.43 million.

The additional NPV of adding shifts if demand is “high” is

$$NPV = -\sum_{t=1}^{10} 29 / 1.09^t = 49.50$$

If demand is low, the production-flexibility option will not be exercised. The optimal decision is to add shifts only if demand is high.

Because the production-flexibility option is exercised only when demand is high, which happens 50% of the time, the expected present value of adding shifts is

$$NPV = 0.50(49.5) = \$24.75 \text{ million.}$$

The total NPV of the initial project and the production-flexibility option is $NPV = -2.43 \text{ million} + 24.75 \text{ million} = \22.32 million.

The option to add shifts, handled optimally, makes this a positive-NPV project. *Uses of Capital*

Question 21

Which of the following institutional investors will most likely have the highest level of risk tolerance?

- A. Life insurance company.
- B. University endowment.**
- C. Commercial bank.

B is correct. Endowments have a higher risk tolerance, as compared to banks and insurance companies. Portfolio Management Overview; Section 3.3.

Question 22

Fran McGuire is estimating the cost of debt of DCM Corporation as part of her valuation analysis of the corporation. DCM has recently issued an eight-year, 10 percent semi-annual bond for USD960. The bond has a face value of USD1,000. If the marginal tax rate is 38 percent, the after-tax cost of debt is closest to:

- A. 5.4%.
- B. 10.8%.
- C. 6.7%.**

C is correct. The pre-tax cost of debt is the YTM of the bond.

Using the Financial Calculator and making adjustments for the semi-annual payments of coupon. $N = 16$, $PV = -960$, $PMT = 0.05 \times 1000 = 50$, $FV = 1000$, $CPT I/Y = 5.38$, annual yield = $5.38 \times 2 = 10.76\% \approx 10.8\%$. After-tax cost of debt = $10.8\% (1 - 38\%) = 6.696 \approx 6.7\%$. Cost of Capital; Section 3.1.

Question 23

An investment manager has the following information regarding his portfolio's return and volatility as compared to the market:

	Return	Risk
Market	6.35%	10.54%
Portfolio	8.68%	15.73%

Given that the risk-free rate is 5.50%, M2 alpha would be closest to:

- A. 3.9 percent.
- B. 1.3 percent.**
- C. -2.6 percent.

B is correct.

$$M2 \text{ alpha} = (R_p - R_f) * (\sigma_m / \sigma_p) - (R_m - R_f)$$

$$M2 \text{ alpha} = (0.0868 - 0.055) * (0.1054 / 0.1573) - (0.0635 - 0.055)$$

$$M2 \text{ alpha} = 1.28\%. \text{ Portfolio Risk and Return Part II; Section 4.3.}$$

Question 24

The risk that a counterparty will not pay an amount owed on an obligation is most likely known as:

- A. credit risk.**
- B. solvency risk.
- C. market risk.

A is correct. Credit risk is defined as the risk that a counterparty will not pay an amount owed on an obligation. Introduction to Risk Management; Section 4.1.

Question 25

Consider the following income statement of a company:

	\$ (millions)
Revenue	21.8
Variable operating costs	8.9
Fixed operating costs	10.5
Operating income	2.4
Interest	0.9
Taxable income	1.5
Tax	0.8
Net income	0.7

The company's degree of operating leverage is closest to:

- A. 5.4.
- B. 4.7.
- C. 9.1.

A is correct. $DOL = [(21.8 - 8.9) / (21.8 - 8.9 - 10.5)] = 5.375$. Measures of Leverage; Section 3.3.

Question 26

Which of the following questions is *most likely* to be asked to mitigate or prevent availability bias?

- A. "How does new information change my forecast?"
- B. **"Am I buying or selling investments because of a recent market event without doing a thorough analysis?"**
- C. "Is the decision the result of focusing on a net gain or net loss position?"

B is correct. The question, "Am I buying or selling investments because of a recent market event without doing a thorough analysis?", would be most likely asked to mitigate availability bias. The question, "Is the decision the result of focusing on a net gain or net loss position?", would be most likely asked to detect and overcome framing bias. The question, "How does new information change my forecast?", would be most likely asked to detect and overcome conservatism bias. *The Behavioral Biases of Individuals*

Question 27

Revolvers are:

- A. in effect for multiple years
- B. often used for much larger amounts
- C. **Both A) and B)**

C is correct. Revolvers differ from regular lines in two ways (1) they are in effect for multiple years and (2) they are often used for much larger amounts. They are the most reliable form of short-term bank borrowing. *Sources of Capital*

Question 28

Samantha purchases stock on margin and holds the position for exactly one year.

Consider the following information:

- Shares purchased: 500
- Purchase price: \$10 per share
- Call money rate: 7%
- Dividend: USD0.5 per share
- Leverage ratio: 2.5
- Total return on equity: 40%

If the interest on loan and dividend are both paid at the end of the year, the price at which Samantha sold the stock is closest to:

- A. \$4.50.
- B. \$9.63.
- C. **\$11.52.**

C is correct.

Total purchase value = $\text{USD}10 \times 500 = \text{USD}5,000$.

Initial equity = Total purchase value / leverage ratio = $\text{USD}5,000 / 2.5 = \text{USD}2,000$.

Amount borrowed = total purchase value – initial equity = $\text{USD}5,000 - \text{USD}2,000 = \text{USD}3,000$.

Margin interest paid = call money rate * amount borrowed = $7\% \times \text{USD}3,000 = \text{USD}210$.

Dividend income = dividend per share * shares purchased = $\text{USD}0.5 \times 500 = \text{USD}250$.

Total return on the initial equity = $40\% \times \text{USD}2,000 = \text{USD}800$.

Gain from price appreciation = total return – dividend + margin interest = $\text{USD}800 - \text{USD}250 + \text{USD}210 = \text{USD}760$.

Price at which the stock was sold = gain from price appreciation per share + purchase price = $(\text{USD}760 / 500 \text{ shares}) + \text{USD}10 = \text{USD}11.52$. Market Organization and Structure; Section 5.2.

Question 29

An analyst gathers the following information about three stocks in a price-return index:

Stock	Shares Outstanding	% Shares in Market Float	Beginning of Period Price (\$)	End of Period Price (\$)	Dividends Per Share (\$)
A	2,500	75	40	55	4
B	3,300	80	60	54	6
C	2,800	55	55	52	5

If the beginning value of the float-adjusted market-capitalization-weighted equity index is 100, the ending value is closest to:

- A. 100.1.
- B. 97.6.
- C. 102.4.

C is correct. This is a price return index (not a total return index). Hence, we only consider changes in prices and ignore the dividends. In float-adjusted market-capitalization weighting, the weight on each constituent security is determined by adjusting its market capitalization for its market float.

Stock	Shares Outstanding	% Shares in Market Float	Beginning of Period Price (USD)	Beg. Float Adjusted Market Cap (USD)	End of Period Price (USD)	Ending Float Adjusted Market Cap (\$)
A	2,500	75	40	75,000	55	103,125
B	3,300	80	60	158,400	54	142,560
C	2,800	55	55	84,700	52	80,080
Total	318,100			325,765		
Index Value	100.0					102.4

$(1) (2) (1) \times (2) = (3) (4) (3) \times (4) = (5) (6) (3) \times (6)$

A 2,500 75 1,875 40 75,000 55 103,125

B 3,300 80 2,640 60 158,400 54 142,560

C 2,800 55 1,540 55 84,700 52 80,080

Total 318,100 325,765

Index Value 100.0 102.4

As per the computations shown above, the ending value of the index equals $325,765 / 318,100 = 102.4$. Security Market Indices; Section 3.2.

Question 30

Which of the following actions taken by an issuer would most likely be considered as a potential breach of a negative covenant?

- A. The issuer decides not to sell its assets in order to finance its own operating activities.
- B. The issuer announces a buy-back of its own shares from the market to reduce its outstanding capital.**
- C. The issuer raises additional debt, which is lower in seniority as compared to the outstanding debt to finance its own operating activities.

B is correct. The purpose of negative covenants is to protect bondholders from such problems as the dilution of their claims, asset withdrawals or substitutions, and suboptimal investments by the issuer. Restrictions on distributions to shareholders which restrict dividends and other payments to shareholders such as share buy-backs (repurchases) could be considered as an action that would reduce the issuer's ability to service outstanding debt. Fixed Income Securities – Defining Elements; Section 3.1.

Question 31

According to the industry life-cycle model, an industry experiencing slowing growth and intense competition is most likely in its:

- A. embryonic stage.
- B. maturity stage.
- C. shakeout stage.**

C is correct. The shakeout stage is usually characterized by slowing growth, intense competition, and declining profitability. During the shakeout stage, demand approaches market saturation levels because few new customers are left to enter the market. Competition is intense as growth becomes increasingly dependent on market share gains. Introduction to Industry and Company Analysis; Section 5.1.

Question 32

Which of the following bonds would appreciate the most if the yield curve shifts down by 100 basis points at all maturities?

- A. 5-year 6% coupon and 7% YTM.
- B. 6-year 6% coupon and 7% YTM.**
- C. 6-year 7% coupon and 7% YTM.

B is correct. When the market discount rates change by the same amount, a longer-term bond has a greater percentage price change than a shorter-term bond (the maturity effect) and a lower-coupon bond has a greater percentage price change than a higher-coupon bond (the coupon effect). Understanding Fixed-Income Risk and Return; Section 3.

Question 33

Which of the following statements about investing is most likely accurate?

- A. An investor who takes a short call position benefits from an increase in the price of the underlying.
- B. An investor who has the right under a contract to sell an underlying asset is said to have a long call position.

C. The return of a borrowed security is referred to as covering the short call position.

C is correct. The return of a borrowed security is referred to as covering a short call position
Statements A and B are not correct. Market Organization and Structure; Section 5.1.

Question 34

Using the following US Treasury spot rates, the arbitrage-free value of a two-year USD100 par value Treasury bond with a 4% semi-annual coupon rate is closest to:

Time Period	Spot Rate	Years
1	1.00%	0.5
2	1.50%	1.0
3	2.00%	1.5
4	2.25%	2.0

- A. \$99.12.
- B. \$100.00.
- C. \$103.44.**

C is correct. The value of the bond is:
 $2 / (1 + .01/2)^1 + 2 / (1 + .015/2)^2 + 2 / (1 + .02/2)^3 + 102 / (1 + .0225/2)^4 = 103.44$. Introduction to Fixed Income Valuation; Section 2.4.

Question 35

An analyst gathers the following information for NYSE-3, a price-weighted index, comprised of Cola, Fizz and Scotch:

Security	Number of shares	Beginning of period price (\$)	End of period price (\$)	Total Dividend (\$)
Cola	1,000	50	49	15
Fizz	1,500	64	72	9
Scotch	1,250	55	68	12

The price return of the index is closest to:

- A. 4.3 percent
- B. 11.8 percent.**
- C. 12.7 percent.

B is correct. The price return of the price-weighted index is the percentage change in price of the index. $(189 - 169) / 169 = 11.83$ percent.

Security	Beginning of period price (\$)	End of period price (\$)
Cola	50	49

Fizz	64	72
Scotch	55	68
Total	169	189

Security Market Indices; Section 2.

Question 36

Which of the following is least likely a characteristic of the shakeout stage?

- A. Increasing profitability.**
- B. Intense competition.
- C. Slow growth.

A is correct. Profitability declines in the shakeout stage due to overcapacity. Introduction to Industry and Company Analysis; Section 5.1.

Question 37

Analyst 1: An arbitrage opportunity is an opportunity to buy an asset at less than its fundamental value.

Analyst 2: An arbitrage opportunity is an opportunity to make a profit at no risk with no capital invested.

- A. Analyst 1 is correct.
- B. Analyst 2 is correct.**
- C. Both analysts are incorrect.

B is correct. An arbitrage opportunity is an opportunity to make a profit at no risk with no capital invested. Derivative Markets and Instruments; Section 7.2.

Question 38

A trader places a stop-buy order. The most likely reason for that could be that the trader:

- A. thinks that the stock is overvalued.
- B. wants to limit the loss on a long position.
- C. wants to limit the loss on a short position.**

C is correct. Investors who have entered into a short sale will incur losses if the stock begins to increase in value. A stop-buy order helps limit the loss on a short position because it becomes valid when the stock price rises above the specified stop price. Market Organization and Structure; Section 6.2.

Question 39

Which of the following statements about callable bonds is most likely correct?

- A. The issuer benefits from lower borrowing costs by issuing callable bonds as compared to similar non-callable bonds.
- B. The issuer can substitute its existing debt with lower-cost debt by calling the bonds.**
- C. The issuer can take advantage of higher interest rates by calling the bonds.

B is correct. Callable bonds give issuers the ability to retire debt prior to maturity. The most compelling reason for them to do so is to take advantage of lower borrowing rates. By calling the

bonds before their maturity date, the issuer can substitute a new, lower cost bond issue for an older, higher cost one. Fixed Income Securities – Defining Elements; Section 5.1.

Question 40

According to put-call parity, which of the following combinations will result in a risk-free position?

- A. Long asset, long put, short call.**
- B. Long call, long put, short asset.
- C. Long asset, long call, short bond.

A is correct. The combination of a long asset, long put, and short call is risk free because its payoffs produce a known cash flow of the value of the exercise price. Basics of Derivative Pricing and Valuation; Section 4.1.

Question 41

According to behavioral finance, the investor bias in which investors dislike losses more than they like comparable gains is best described as:

- A. loss aversion.**
- B. overconfidence.
- C. risk aversion.

A is correct. Loss aversion is where investors dislike losses more than they like comparable gains. Market Efficiency; Section 5.

Question 42

Which of the following industries is *most likely* to be characterized as concentrated with weak pricing power?

- A. Integrated Oil**
- B. Home Improvement
- C. Restaurants

A is correct. Integrated Oil is an example of concentrated industry with weak pricing power. The home improvement industry is fragmented and possesses strong pricing power. Restaurants is an example of fragmented industry with weak pricing power. *Industry and Company Analysis*

Question 43

In credit analysis, relying only on credit rating agency's ratings has limitations because:

- A. of huge growth of debt markets.
- B. of independent assessment of credit risk.
- C. credit spreads move more quickly than rating agencies change ratings.**

C is correct. Credit ratings tend to lag the market's pricing of credit risk. Credit spreads and bond prices move more quickly because of changes in perceived creditworthiness than rating agencies change their ratings up or down. Bond prices and relative valuations can move every day, whereas bond ratings, appropriately, don't change that often. Fundamentals of Credit Analysis; Section 4.3.

Question 44

In case of capital indexed bonds linked to inflation, when inflation increases:

- A. the principal amount remains unchanged but the coupon rate increases.
- B. the coupon rate remains unchanged but the principal amount increases.**
- C. the coupon payment remains unchanged but the principal amount increases.

B is correct. Following an increase in inflation, the coupon rate of a capital-indexed bond remains unchanged, but the principal amount is adjusted upward for inflation. Thus, the coupon payment, which is equal to the fixed coupon rate multiplied by the inflation-adjusted principal amount, increases. Fixed Income Securities – Defining Elements; Section 2.1.

Question 45

Momentum anomalies most likely:

- A. relate to long-term price patterns.
- B. are consistent with weak-form market efficiency.
- C. are caused by investor overreaction.**

C is correct. Momentum anomalies result from investor overreaction in response to the release of unexpected public information. Market Efficiency; Section 4.1.

Question 46

Which of the following statements about repurchase agreements and reverse repurchase agreements is most likely accurate?

- A. Reverse repurchase agreements are often used to cover short positions.**
- B. In a repurchase agreement, the repurchase price is determined on the repurchase date.
- C. If the dealer is borrowing a security and lending cash to the counterparty, from the dealer's perspective the transaction is termed as a repurchase agreement.

A is correct. In a reverse repurchase agreement, the dealer is borrowing a security and lending cash to the counterparty. The transaction can be structured by the dealer to borrow a security to cover a short position in the market. B is incorrect because in a repurchase agreement the repurchase price is determined on the same day and not the repurchase date. C is incorrect because in a repurchase agreement, the dealer is borrowing cash and lending a security to the counterparty. Fixed Income Markets – Issuance Trading and Funding; Section 8.3.

Question 47

Assuming a floating-rate note in a market exists with a quoted margin of +25 basis points and a required margin of +40 basis points, the price of the note on its next reset date will most likely be:

- A. less than par value.**
- B. equal to par value.
- C. greater than par value.

A is correct. If the required margin is greater than the quoted margin, the credit quality of the issue has decreased and the price on the next reset date will be less than par value. Introduction to Fixed Income Valuation; Section 3.4.

Question 48

Which of the following statements about forward rate agreements is/are most likely correct?

Statement I: A bank that intends to lend in the future can hedge its interest rate risk by taking a short position in a forward rate agreement.

Statement II: A company that expects to borrow in the future can hedge its interest rate risk by taking a long position in a forward rate agreement.

- A. Statement I but not Statement II.
- B. Statement I and II.**
- C. Neither Statement I nor Statement II.

B is correct. Statement I is correct because an entity that wishes to lend in the future can in a way lock-in the rate that is the price of the forward rate agreement at initiation thereby hedging the interest rate risk. In case of a decline in rates, the return on the funds loaned at the future date would be lower but a positive payoff on the forward rate agreement would compensate these returns. Statement II is based on the same concept but in this case the position is of a borrower and hence a long position in the forward rate agreement would hedge interest rate risk. Basics of Derivative Pricing and Valuation; Section 3.1.

Question 49

Ms. Yang wants to sell a large block of an illiquid stock from her portfolio and seeks advice from her trader. The trader will most likely execute the trade in a(n):

- A. brokered market.**
- B. order-driven market.
- C. quote-driven market.

A is correct. Instruments that are infrequently traded and expensive to carry as inventory (e.g. very large blocks of stock, real estate) are executed in brokered markets. Organizing order-driven markets for such instruments is not sensible because too few traders would submit orders for them. Market Organization and Structure; Section 8.2.

Question 50

The principal balance of a pool is USD10,000,000 and USD100,000 is scheduled to be repaid in a given month. The SMM is 0.89%. The forecasted prepayment amount for the month is closest to:

- A. 890.
- B. 88,110.**
- C. 89,000.

B is correct. $\text{Prepayment} = \text{SMM} * (\text{Beginning balance} - \text{Scheduled principal repayment}) = 0.89\% * (10,000,000 - 100,000) = 88,110$. Introduction to Asset Backed Securities; Section 5.1.

Question 51

An investor who would like to get public market debt exposure to real estate would most likely consider investing in:

- A. collateralized mortgage obligations.**
- B. shares of real estate investment trusts.
- C. direct mortgage lending.

A is correct. Collateralized mortgage obligation is a form of instrument that is publicly traded and gives debt exposure to real estate investing. Shares of real estate investment trusts is an example of public market equity exposure and direct mortgage lending is an example of private debt exposure to real estate investing. Introduction to Alternative Investments; Section 5.1.

Question 52

An investor who wants to exploit the value effect is most likely to purchase shares of a company which has higher:

- A. market-to-book ratios.
- B. dividend yields.**
- C. price-to-earnings ratios.

B is correct. Value stocks have higher dividend yields than average, and lower price-to-earnings and market-to-book ratios. Equity Valuation; Section 4.2.

Question 53

Which of the following is most likely correct with respect to Commodity indexes?

- A. Because the commodity indexes consist of futures contracts on the commodities rather than the actual commodities, the return of the commodity index reflects changes in future prices only.
- B. The return of all the commodity indexes is different because each index may use a different weighting method.**
- C. Commodity indexes that target the same market share similar risk and return profiles.

B is correct.

Although some commodity indexes may include the same commodities, the returns of these indexes may differ because each index may use a different weighting method.

Commodity index returns reflect the risk-free interest rate, the changes in future prices, and the roll yield.

Unlike commodity indexes, broad equity and fixed-income indexes that target the same markets share similar risk and return profiles. Security Market Indexes. Section 7.1. LO. j.

Question 54

Which of the following statements about exchange-traded derivatives is most accurate?

- A. They are less transparent than over-the-counter derivatives.
- B. All terms of the contract except the price are standardized.**
- C. They have more credit risk than over-the-counter derivatives.

B is correct. Exchange-traded contracts are standardized, meaning that the exchange determines the terms of the contract except the price. Derivative Markets and Instruments; Section 3.1.

Question 55

A 6-year annual pay corporate bond with a 3.25% coupon is priced at \$98.36. The bond's duration is 5.8 and its convexity is 36. Assuming the bond's credit spread narrows by 75 basis points, the approximate return impact on the bond considering its convexity is closest to:

- A. -4.45%.
- B. 5.80%.
- C. 4.45%.**

C is correct. Return impact $\approx -\text{modified duration} \times \Delta\text{spread} + 1/2 \text{ convexity} \times [(\Delta\text{spread})]^2$.

Return impact $\approx -5.8 \times -0.0075 + 1/2 (36.0) \times [(-0.0075)]^2$. Return impact $\approx 4.45\%$.

Understanding Fixed-Income Risk and Return; Section 3.4.

Question 56

A South Korean company that has operations in the UK and the US issues a US dollar denominated bond. The price of this bond is most likely to:

- A. change as UK interest rates change.
- B. change as US interest rates change.**
- C. change as South Korean interest rates change.

B is correct. The currency denomination of a bond's cash flows influences which country's interest rates affect a bond's price. The price of a bond issued by a South Korean company and denominated in US dollars will be affected by US interest rates. Fixed Income Securities – Defining Elements; Section 3.2.

Question 57

The value of a non-dividend company with a high proportion of intangible assets is most appropriately found by using:

- A. Gordon growth model.
- B. asset-based valuation model.
- C. free-cash-flow-to-equity model.**

C is correct. Free-cash-flow-to-equity model is the most appropriate. Gordon growth is unsuitable since the company does not pay dividends; asset-based model is unsuitable due to high proportion of intangibles. Equity Valuation; Section 4.2.

Question 58

On 1 January 2014, Paul Corp. issued USD100 par value, floating rate bonds at 3-month LIBOR + 100 bps. These bonds pay interest on a quarterly basis. We are now at the end of June and the second quarterly interest payment is due. What is the coupon payment bondholders will receive? Assume that: 6-month LIBOR at the beginning of January was 4%, 3-month LIBOR at the end of March was 2% and 3-month LIBOR now is 3.50%?

- A. \$3.00.
- B. \$0.75.**
- C. \$1.25.

B is correct. The applicable interest rate for the second payment due in June is the 3-month LIBOR at the end of March plus 100 bps. Therefore, coupon rate is equal to $(2.00\% + 1.00\%) = 3.00\%$. However, the payments are made quarterly. Interest payment in June = $(3.00\%/4) \times \text{USD}100 = \text{USD}0.75$. Introduction to Fixed Income Valuation; Section 3.4.

Question 59

Which of the following derivatives is a contingent claim?

- A. Credit default swap.**
- B. Interest rate swap.
- C. Forward contracts.

A is correct. A contingent claim is a claim that depends on a particular event and only requires a payment if a certain threshold price is broken. A credit default swap is a contingent claim as the payoff depends on a credit event such as default. Derivative Markets and Instruments; Section 4.2.

Question 60

The following information about Cola Inc. has been gathered for analysis:

ROE	12 percent
Retention Ratio	60 percent.
Required Return on Shares	11 percent.
Next Year's EPS	\$9

The company's justified forward P/E ratio is closest to:

- A. \$15.8x.
- B. \$9.7x.
- C. \$10.5x.**

C is correct. $\text{Growth} = \text{Retention Ratio} * \text{ROE} = 0.6 * 0.12 = 0.072$.

$\text{Justified Forward PE} = (\text{Payout Ratio}) / (r - g) = (1 - 0.6) / (0.11 - 0.072) = 10.53$. Equity Valuation; Section 5.1.

Question 61

Which of the following statements about covered bonds is *least* accurate?

- A. Covered bonds offer bondholders dual recourse.
- B. Covered bonds usually consist of one bond class per cover pool whereas ABS often comprise of several bond classes.
- C. The underlying asset pool is transferred to a separate SPV.**

C is correct. The underlying asset pool remains on the issuing financial institution's balance sheet. It is not transferred to a separate SPV. *Introduction to Asset-Backed Securities*

Question 62

Following data on a company is provided:

Year	EPS	DPS	ROE
2013	4.45	1.85	14%
2012	2.50	0.50	12%
2011	3.20	1.10	13%
2010	2.78	0.85	12%
2009	2.15	0.77	11%
2008	2.05	0.75	10%

Assume the estimated dividend growth rate is the average of the compounded annual growth rate of dividends over the period 2008–2013 and the sustainable growth rate for the year 2013. The required return is 17%. Using Gordon's constant growth dividend discount model, the stock's intrinsic value is closest to:

- A. USD22.7.
- B. USD54.8.
- C. USD70.3.**

C is correct.

Dividend growth rate over the period 2008 to 2013: The dividend in 2008 was 0.75. The dividend in 2013 was 1.85. Let the growth rate be 'g'. We can use the TVM function of the calculator to find the growth rate. $N=5$, $PV = 0.75$, $PMT = 0$, $FV = 1.85$; CPT I/Y, I/Y = 19.8%

Sustainable growth rate for the year 2013: $g = RR \times ROE = (4.45 - 1.85) / 4.45 \times 14\% = 8.2\%$.

(The retention ratio is calculated as earnings retained/total earnings i.e. $(4.45 - 1.85) / 4.45$)

Average = $(19.8 + 8.2) / 2 = 14\%$

$V_0 = D_1 / (r - g) = 1.85 \times 1.14 / (0.17 - 0.14) = 70.3$. (D1 is the next period dividend. We take the 2013 dividend of 1.85 and increase it by the estimated dividend growth rate of 14%. $D_1 = 1.85 \times 1.14$)

Equity Valuation; Section 4.4.

Question 63

The highest-ranked unsecured debt is:

- A. senior unsecured debt.**
- B. senior subordinated debt.
- C. first lien loan.

A is correct. First lien loan is a secured loan that ranks first in both collateral protection and priority of payment. The highest-ranked unsecured debt is senior unsecured debt. Other, lower-ranked debt includes senior subordinated debt and junior subordinated debt.

Fundamentals of Credit Analysis; Section 3.2.

Question 64

Which of the following statements is/are most likely correct?

Statement I: The annual payment of a fully amortized bond can be viewed as an annuity.

Statement II: A sinking fund arrangement reduces the risk the issuer will default when the principal is due, thereby reducing the credit risk of the bond issue.

Statement III: A bond that is fully amortized is characterized by a fixed periodic payment schedule that reduces the bond's outstanding principal amount however, there is a balloon payment at maturity as well.

- A. Statement II only.
- B. Statement I and II.**
- C. Statement I and III.

B is correct. Statement I and II are correct. Statement III is incorrect because a bond that is fully amortized is characterized by a fixed periodic payment schedule that reduces the bond's outstanding principal amount to zero by the maturity date. A partially amortized bond also makes fixed periodic payments until maturity, but only a portion of the principal is repaid by the maturity date. Thus, a balloon payment is required at maturity to retire the bond's outstanding principal amount. Fixed Income Securities – Defining Elements; Section 4.1.

Question 65

Baskit Ltd. is a French company which exports confectionary to the United States. The firm has a policy of converting all foreign currency receipts into euros. Expecting to receive USD85 million from a customer in three months, Baskit's chief financial officer is most likely to hedge the risk of an adverse FX movement by:

- A. buying dollars in the forward market.
- B. selling euros in the forward market.**

C. selling dollars in the forward market.

C is correct. Selling dollars in the forward market locks in a conversion rate and hedges against the risk of the dollar depreciating against the euro. Market Organization and Structure; Section 2.1.

Question 66

Which of the following strategies best replicates a receive variable pay fixed swap?

- A. Purchasing a series of forward contracts, each with an initial value of zero.
- B. Purchasing a floating rate bond financed by issuing a fixed rate bond.**
- C. Purchasing a fixed rate bond financed by issuing a floating rate bond.

B is correct. The payment structure of a receive variable pay fixed swap can be replicated by going long on a floating rate bond and going short on a fixed rate bond.

C is incorrect, because it replicates a receive fixed pay variable swap.

A is incorrect. Due to differences in timing, the series of forward contracts used need to be off-market, i.e. they will have a non-zero value at initiation. Basics of Derivative Pricing and Valuation; Section 3.3.

Question 67

Which of the following is most likely to be directly reflected in the returns on a commodity index?

- A. Roll yield.**
- B. Changes in the spot prices of underlying commodities.
- C. Changes in the returns of underlying commodities.

A is correct. Commodity index returns reflect the changes in future prices and the roll yield. Changes and returns in the underlying commodity spot prices are not reflected in a commodity index. Security Market Indexes; Section 7.1.

Question 68

A US bank issued USD100 million of 90-day, 3% US commercial paper. The bank will pay the investor:

- A. \$100,750,000 at maturity.
- B. \$750,000 at maturity.
- C. \$100,000,000 at maturity.**

C is correct. US commercial paper is typically issued on a discount basis—that is, US commercial paper is issued at a discount to par value and pays full par value at maturity. The difference between the par value and the issue price is the interest paid on the borrowing. Fixed Income Markets – Issuance Trading and Funding; Section 6.2.

Question 69

The management fee for private equity funds is usually based on:

- A. initial investment.
- B. committed capital.**
- C. assets under management.

B is correct. The management fee for private equity funds is based on committed capital whereas for hedge funds the management fees are based on assets under management. Introduction to Alternative Investments; Section 4.1.

Question 70

Which of the following statements on commodity investing is least likely accurate?

- A. If commodity prices determine inflation index levels, over time, commodities would on average yield a zero-real return.
- B. Investors can own shares in a company exposed to commodity such as oil to gain a direct exposure to commodity prices.**
- C. Historically, commodities have exhibited a low correlation with traditional assets.

B is correct. Investors may consider owning shares in a commodity-exposed company in order to have an exposure to commodities. However, this would be considered as indirect exposure as it is unclear that the performance of these stocks closely tracks the performance of the underlying commodity. Introduction to Alternative Investments; Section 6.

Question 71

The flat price is usually quoted by bond dealers because:

- A. flat price is adjusted after each coupon payment.
- B. flat price includes accrued interest which is to be paid by the buyer.
- C. flat price avoids misleading investors about the market price trend for the bond.**

C is correct. The reason for using the flat price for quotation is to avoid misleading investors about the market price trend for the bond. If the full price were to be quoted by dealers, investors would see the price rise day after day even if the yield-to-maturity did not change. That is because the amount of accrued interest increases each day. Then, after the coupon payment is made, the quoted price would drop dramatically. Using the flat price for quotation avoids that misrepresentation. Introduction to Fixed Income Valuation; Section 3.1.

Question 72

Rachel thoroughly studies and analyzes financial information before buying stocks. Her peers rely on her judgement and imitate her investment decisions. This behavioral bias most likely:

- A. illustrates irrational behavior.
- B. improves market efficiency.**
- C. decreases market efficiency.

B is correct. The behavior of informed traders acting first and uninformed traders imitating the informed traders is consistent with rationality. The imitation trading by the uninformed traders helps the market incorporate relevant information and improves market efficiency. Market Efficiency; Section 2.3.

Question 73

A call option is most likely to be more valuable if:

- A. the underlying asset has a convenience yield.
- B. the underlying asset pays a dividend.
- C. storage costs of the underlying asset are positive.**

C is correct. Positive storage costs make holding an asset more costly. This makes holding a call option more valuable because call holders can have long exposure to the asset without paying the costs of actually owning the asset. If there are benefits of holding the underlying assets such as dividends or a convenience yield, it decreases the value of call options. *Derivative Markets and Instruments; Section 4.2.*

Question 74

The structure of a sequential pay CMO allows contraction risk:

- A. to be completely eliminated; however, extension risk still exists.
- B. and extension risk to be redistributed between the tranches.**
- C. to be redistributed; however, extension risk is completely eliminated.

B is correct. With the sequential pay CMO structure, contraction and extension risk still exist but they have been redistributed between the tranches. The shorter tranche which matures first offers more protection against extension risk and the longer tranche offers relatively more protection against contraction risk. *Introduction to Asset Backed Securities; Section 5.1.*

Question 75

Which of the following statements about empirical and analytical durations is *least* accurate?

- A. Empirical duration uses statistical methods and historical bond prices to derive the price-yield relationship for specific bonds or bond portfolios.
- B. Analytical duration implicitly assumes that benchmark yields and credit spreads are uncorrelated with one another.
- C. For a government bonds with little or no credit risk, the empirical duration will be lower than their analytical duration.**

C is correct. For a government bond with little or no credit risk, the analytical and empirical duration would be similar because bond prices are largely driven by changes in the benchmark yield. *Understanding Fixed-Income Risk and Return*

Question 76

Which of the following metrics used to review the performance of alternative investments is a variation of the Calmar ratio?

- A. Sortino ratio
- B. MAR ratio**
- C. Batting average

B is correct. Calmar ratio is average return relative to the worst drawdown loss (distance between a peak and a trough of a portfolio). It is typically calculated using the prior three years of data. MAR ratio is a variation of the Calmar ratio. Instead of just three years, it uses the full investment history and the average drawdown.

Batting average refers to the percentage of profitable trades.

Sortino ratio is a measure of return relative to downside volatility. *Introduction to Alternative Investments*

Question 77

A 10-year annual pay bond with 8% coupon and face value of USD1,000 is currently priced at USD952.52. Duration of the bond is 9.52 and convexity is 23.65. Recently due to weakening economy the bond's credit spread widened by 50 basis points. The bond's price will most likely:

- A. decrease by 4.73%.**
- B. decrease by 4.93%.
- C. increase by 4.71%.

A is correct. As the yield widens, the bond's price will fall.

Price impact = $- \text{modified duration} * \Delta \text{spread} + (1/2) * \text{convexity} * (\Delta \text{spread})^2$

Price impact = $-9.52 * (0.0050) + (1/2) (23.65) * (0.0050)^2 = -0.0476 + 0.000296 = -0.0473 = -4.73\%$

Fundamentals of Credit Analysis; Section 6.

Question 78

Which of the following is not a contingent claim?

- A. A put option.
- B. An interest rate swap.**
- C. A credit default swap.

B is correct. An interest rate swap is a forward commitment not a contingent claim. Derivative Markets and Instruments; Section 4.2.

Question 79

Which of the following statements about swaps is most likely correct?

- A. Swaps typically trade on an exchange.
- B. Swaps are largely unregulated.**
- C. Swaps require a daily settlement of gains and losses.

B is correct. Typically, swaps do not trade in any organized secondary market and are largely unregulated. Swap payments are made on periodic settlement dates as specified in the contract and do not require daily settlement of gains and losses. Derivative Markets and Instruments; Section 4.1.

Question 80

The value of a forward contract prior to expiration is the value of the asset:

- A. minus the forward price.
- B. minus the present value of the forward price.**
- C. plus the present value of the forward price.

B is correct. The value of a forward contract prior to expiration is the value of the asset minus the present value of the forward price. Basics of Derivative Pricing and Valuation; Section 3.1.

Question 81

Which of the following statements about venture capital investments is most likely accurate?

- A. Angel investing refers to capital provided at product development stage.
- B. Mezzanine venture capital investing refers to capital provided to take the company public.**
- C. Venture capitalists are passive investors.

B is correct. Angel investing is capital provided at the idea stage. Funds may be used to transform the idea into a business plan and to assess market potential. Venture capitalists are not passive investors. They are actively involved with the companies in which they invest. Introduction to Alternative Investments; Section 4.2

Question 82

Assume that the GP has earned an 19% IRR on an investment, the hurdle rate is 8%, and the partnership agreement includes a catch-up clause. Which of the following statements is *most* accurate about profit distribution in this case?

- A. **The GP effectively earns 3.8% and the LP effectively earns 15.20%.**
- B. The GP effectively earns 2.2% and the LP effectively earns 16.8%.
- C. The GP effectively earns 2% and the LP effectively earns 17%.

A is correct.

- The LPs would receive the entirety of the first 8% profit.
- The GP would receive the entirety of the next 2% profit—because 2% out of 10% amounts to 20% of the profits accounted for so far.
- The remaining 9% would be split 80/20 between the LPs and the GP.
- Thus, the GP effectively earns: $19\% \times 20\% = 3.8\%$ and the LP effectively earns $19\% \times 80\% = 15.2\%$. *Introduction to Alternative Investments*

Question 83

Which of the following statements is not accurate about waterfall?

- A. It defines the way in which cash distributions will be allocated between the GP and the LPs.
- B. In most waterfalls, a GP receives a disproportionately larger share of the total profits relative to their initial investment.
- C. **European waterfalls are more advantageous for a GP compared to American waterfalls.**

C is correct. In case of deal-by-deal (or American) waterfalls, performance fees are collected on a per-deal basis. This is more advantageous for a GP as he can get paid before LPs receive both their initial investment and their preferred rate of return on the entire fund.

In case of whole-of-fund (or European) waterfalls, as deals are exited, all distributions go to the LPs first. The GP does not participate in any profits until the LPs receive their initial investment and the hurdle rate has been met. *Introduction to Alternative Investments*

Question 84

Consider an American call and a European call on the same underlying asset with the same strike price and expiration date. Which of the following is least likely?

- A. Both American call and European call priced at USD5.
- B. **American call priced at USD4 and European call priced at USD5.**
- C. American call priced at USD5 and European call priced at USD4.

B is correct. The American options possess every characteristic of European options and one additional trait: they can be exercised at any time prior to expiration. Early exercise cannot be required, so the right to exercise early cannot have negative value. Thus, American options cannot sell for less than European options. Basics of Derivative Pricing and Valuation; Section 4.3.

Question 85

Which type of a real estate index is most likely to exhibit sample selection bias?

- A. Appraisal index.
- B. REIT index.
- C. Repeat sales index.**

C is correct. A repeat sales index includes prices of properties that have been sold recently. As these properties may not be representative of the overall property market, there is a risk of sample selection bias. In the case of appraisal index and a REIT index, a sample of representative properties or REIT property pools are considered. Introduction to Alternative Investments; Section 5.3.

Question 86

Pinks Inc. uses cash to buy back common shares. The most likely effect of this will be a:

- A. higher growth rate.
- B. higher profits.
- C. higher ROE.**

C is correct. The growth rate and profits will probably not change much. The equity will decrease. Hence, ROE will increase. Equity Valuation; Section 4.1.

Question 87

The characteristics that most likely make a company attractive as a leveraged buyout target are:

- A. a company with a strong cash flow but the company is currently managed inefficiently.**
- B. a company that is inefficiently managed and its stock price is overvalued.
- C. a company that has a high amount of debt on its balance sheet and a management that is looking for a buyout deal.

A is correct. The characteristics that make a company attractive as an LBO target are 1) undervalued stock price 2) existing management that is looking for a deal 3) inefficient companies that have the potential to perform well if managed better 4) strong and sustainable cash flows 5) low leverage and 6) high amount of physical assets. Introduction to Alternative Investments; Section 4.2.

Question 88

Which of the following is most likely an example of a derivative?

- A. A mutual fund.
- B. An exchange traded fund.
- C. A forward rate agreement.**

C is correct. A forward rate agreement transforms the performance of the underlying interest rate and is an example of a derivative. A and B are incorrect. Although mutual funds and exchange traded funds derive their value from the underlying assets they hold, they do not transform the performance of the underlying assets and hence are not derivatives. Derivative Markets and Instruments; Section 2.

Question 89

Which type of alternative investment approaches is *most likely* subject to adverse selection bias?

- A. Fund investing
- B. Co-investing**
- C. Direct investing

B is correct. Co-investing may be subject to adverse selection bias. For example, the fund manager may direct less attractive investment opportunities to the co-investor while allocating its own capital to more lucrative deals. *Introduction to Alternative Investments*

Question 90

SHM Capital is a hedge fund with USD200 million of initial investment capital. They charge a 3 percent management fee based on assets under management at year-end and a 15 percent incentive fee. In its first year, SHM Capital has a 28 percent return. Assume management fees are calculated using end-of-period valuation. In the second year, the fund value declines to USD225 million. In the third year, the fund value increases to USD250 million. If the incentive fee is calculated net of the management fee, and also includes the use of a high-water mark, the fees earned by SHM in the second year is *closest* to:

- A. USD2.10 million.
- B. USD4.65 million.
- C. USD6.75 million.**

C is correct. Management fee at the end of first year = $200 \text{ million} * 1.28 * 0.03 = \text{USD}7.68$ million. Incentive fee at the end of first year = $((200 \text{ million} * 1.28) - 200 - 7.68) * 0.15 = \text{USD}7.248$ million

Beginning capital position at the start of second year = $(200 \text{ million} * 1.28) - 7.68 - 7.248 = \text{USD}241.072$ million. USD241.072 million represents the high-water mark established at the end of Year 1. Management fee at the end of second year = $225 * 0.03 = \text{USD}6.75$ million. No incentive fee because the fund has declined in value. Total fees to SHM Capital = USD6.75 million. *Introduction to Alternative Investments; Section 3.3. LO.e.*